

Straight lines - Set 1 - Solutions

Specification MM3.1, M4.9, M4.10.

Answer key

Question	Answer	Topic
1	C	Straight lines
2	F	Straight lines
3	A	Straight lines
4	A	Straight lines
5	B	Straight lines
6	A	Straight lines
7	H	Straight lines
8	D	Straight lines
9	B	Straight lines
10	A	Straight lines
11	B	Straight lines
12	F	Straight lines

Worked solutions

Question 1 LIN-01 • Straight lines • spec MM3.1

Find the equation of the line through $(2, -3)$ that is perpendicular to $3x - 2y + 5 = 0$.

Answer: C

The given line has gradient $\frac{3}{2}$, so the perpendicular has gradient $-\frac{2}{3}$:

$$y + 3 = -\frac{2}{3}(x - 2) \Rightarrow 3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0.$$

Full method - how and why

Perpendicular line through a point: find the gradient of the given line, take the negative reciprocal, then use the point-gradient form.

- 1 Rearrange $3x - 2y + 5 = 0$ into $y = \frac{3}{2}x + \frac{5}{2}$, so its gradient is $\frac{3}{2}$.
why: Alternatively use the shortcut: $ax + by + c = 0$ has gradient $-\frac{a}{b} = -\frac{3}{-2} = \frac{3}{2}$.
- 2 Perpendicular gradient: $-\frac{1}{3/2} = -\frac{2}{3}$.
why: Perpendicular gradients multiply to -1 ; invert and change sign.
- 3 Use $y - y_1 = m(x - x_1)$ at $(2, -3)$: $y + 3 = -\frac{2}{3}(x - 2)$.
why: $y - (-3)$ is $y + 3$ - the commonest sign slip in this whole topic.
- 4 Clear fractions and tidy: $3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0$.
why: Multiply through by 3 first, then collect everything on one side.

Where it goes wrong

- Using $y - 3$ instead of $y + 3$, which gives $2x + 3y - 5 = 0$ - an offered option.
- Taking the perpendicular gradient as $-\frac{3}{2}$ or $\frac{2}{3}$.

Takeaway Negative **reciprocal**, and substitute the point with its own sign: $y - (-3) = y + 3$.

Question 2 LIN-02 • Straight lines • spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

$$\text{Gradient } AB = \frac{2}{4} = \frac{1}{2}. \text{ Gradient } AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k - 1 = 12 \Rightarrow k = 13.$$

Full method - how and why

Collinear points have equal gradients between any pair. Set two gradients equal and solve.

- 1 Gradient of AB : $\frac{4-2}{5-1} = \frac{2}{4} = \frac{1}{2}$.
why: Any two of the three points determine the line, so compute the gradient from the pair with known coordinates.
- 2 Gradient of AC : $\frac{8-2}{k-1} = \frac{6}{k-1}$.
why: Use the same starting point A for both gradients to avoid sign confusion.
- 3 Set them equal: $\frac{6}{k-1} = \frac{1}{2}$.
why: Collinear means the gradient between any pair is the same.
- 4 Cross-multiply: $k - 1 = 12$, so $k = 13$.
why: Check: from $B(5, 4)$ to $C(13, 8)$ the gradient is $\frac{4}{8} = \frac{1}{2}$ ✓.

Where it goes wrong

- Forgetting the -1 and answering 12.
- Inverting the gradient and solving $\frac{k-1}{6} = \frac{1}{2}$, which gives $k = 4$.

Takeaway Collinearity is an equal-gradient condition; always verify with the third pair of points.

Question 3 LIN-05 · Straight lines · spec MM3.1

Which of the following lines is parallel to $4x + 6y = 9$?

Answer: A

$4x + 6y = 9$ has gradient $-\frac{2}{3}$; so does $2x + 3y = 1$.

Full method - how and why

Parallel means equal gradient. Convert each option to gradient form - or compare the coefficient ratios directly.

1 Gradient of $4x + 6y = 9$ is $-\frac{4}{6} = -\frac{2}{3}$.

why: For $ax + by = c$ the gradient is $-\frac{a}{b}$.

2 Test the options: $2x + 3y = 1$ has gradient $-\frac{2}{3}$ ✓.

why: Equivalently, the coefficients (2, 3) are proportional to (4, 6) - the constant term is irrelevant to direction.

3 Reject the rest: $3x - 2y = 4$ gives $\frac{3}{2}$ (perpendicular), $4x - 6y = 9$ gives $\frac{2}{3}$, $6x + 4y = 9$ gives $-\frac{3}{2}$, and $y = \frac{3}{2}x + 1$ gives $\frac{3}{2}$.

why: Note how many distractors are the **perpendicular** gradient - a deliberate trap.

Where it goes wrong

- Choosing $6x + 4y = 9$ because the numbers 4 and 6 reappear - but swapped, so the gradient changes.
- Forgetting the minus sign in $-\frac{a}{b}$.

Takeaway Parallel $\Leftrightarrow$ the coefficients of x and y are in the same ratio; the constant plays no part.

Question 4 LIN-06 · Straight lines · spec MM3.1

The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

Answer: A

Midpoint (4, 3); gradient of segment $\frac{-4}{6} = -\frac{2}{3}$, so the bisector has gradient $\frac{3}{2}$:
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

Full method - how and why

Perpendicular bisector: through the midpoint, perpendicular to the segment. Two small calculations, then the point-gradient form.

1 Midpoint: $(\frac{1+7}{2}, \frac{5+1}{2}) = (4, 3)$.

why: The bisector must pass through the middle of the segment.

2 Gradient of the segment: $\frac{1-5}{7-1} = -\frac{4}{6} = -\frac{2}{3}$.

why: Keep the order of subtraction consistent in numerator and denominator.

3 Perpendicular gradient: $\frac{3}{2}$.

why: Negative reciprocal of $-\frac{2}{3}$.

4 Line through (4, 3): $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

why: Multiply by 2 to clear the fraction, then collect.

Where it goes wrong

- Rearranging to $3x - 2y = 18$ by a sign slip on the constant.
- Using the segment's gradient instead of the perpendicular one.

Takeaway Perpendicular bisector = midpoint + negative reciprocal gradient; it is the locus of points equidistant from the two ends.

Question 5 LIN-07 · Straight lines · spec MM3.1

The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.

Answer: B

$$(3, 0): 3a = 12 \Rightarrow a = 4. (0, -4): -4b = 12 \Rightarrow b = -3. a - b = 7.$$

Full method - how and why

Intercepts are points on the line. Substitute each one to get an equation for a and for b separately.

- 1 x -intercept 3 means the point $(3, 0)$ lies on the line: $3a + 0 = 12$, so $a = 4$.

why: At an x -intercept, $y = 0$.

- 2 y -intercept -4 means $(0, -4)$ lies on it: $0 - 4b = 12$, so $b = -3$.

why: At a y -intercept, $x = 0$. The negative intercept makes b negative.

- 3 Compute $a - b = 4 - (-3) = 7$.

why: Subtracting a negative adds.

Where it goes wrong

- Writing $b = 3$ and answering 1.
- Mixing up which intercept is which and substituting $(0, 3)$ and $(-4, 0)$.

Takeaway An intercept is just a point with one coordinate zero - substitute it into the equation.

Question 6 LIN-14 · Straight lines · spec MM3.1

A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

Answer: A

$$\text{Midpoint } (1, 2) \text{ means } A = (2, 0) \text{ and } B = (0, 4). \text{ Gradient} = \frac{4-0}{0-2} = -2.$$

Full method - how and why

The midpoint condition determines both intercepts, and two points give the gradient.

- 1 Let $A = (p, 0)$ and $B = (0, q)$. Their midpoint is $(\frac{p}{2}, \frac{q}{2})$.

why: One intercept on each axis, so each has a zero coordinate.

- 2 Set the midpoint equal to $(1, 2)$: $\frac{p}{2} = 1$ and $\frac{q}{2} = 2$, so $p = 2$, $q = 4$.

why: Equate coordinates separately.

- 3 Gradient through $(2, 0)$ and $(0, 4)$: $\frac{4-0}{0-2} = -2$.

why: Consistent order of subtraction in numerator and denominator.

Where it goes wrong

- Halving instead of doubling, giving $A = (\frac{1}{2}, 0)$ and the wrong gradient.
- Answering 2 by ignoring that the line falls from left to right.

Takeaway If $(1, 2)$ is the midpoint of the intercepts, the intercepts are $(2, 0)$ and $(0, 4)$ - double each coordinate.

Question 7 X2-05 · Straight lines · spec MM3.1

Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II If $m_1 = m_2$ then the lines do not meet.
- III If the lines meet on the y -axis then $c_1 = c_2$.

Answer: H

I: perpendicular lines are not parallel, so they meet once. II: equal gradients and distinct lines means parallel and different, so no intersection. III: on the y -axis $x = 0$, so $y = c_1 = c_2$.

Full method - how and why

Three claims about two distinct straight lines. Each follows from what gradients and intercepts mean.

- 1 I: $m_1m_2 = -1$ means the lines are perpendicular, so $m_1 \neq m_2$ and the lines are not parallel - they meet exactly once. **True.**
why: Two non-parallel straight lines always meet at exactly one point.
- 2 II: $m_1 = m_2$ with the lines **distinct** means $c_1 \neq c_2$: parallel and separate, so they never meet. **True.**
why: The word 'distinct' in the question is doing real work - without it the lines could coincide.
- 3 III: on the y -axis $x = 0$, so the two y -values are c_1 and c_2 . Meeting there forces $c_1 = c_2$. **True.**
why: Substituting $x = 0$ into both equations.
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Rejecting II by imagining the lines could coincide - 'distinct' rules that out.
- Rejecting III by thinking they might meet elsewhere; the claim conditions on them meeting ****on the y -axis****.

Takeaway Same gradient and distinct $\Rightarrow$ parallel, never meet. Different gradients $\Rightarrow$ exactly one intersection. Read 'distinct' carefully.

Question 8 LIN-03 · Straight lines · spec MM3.1

The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

Answer: D

Area of triangle $= \frac{1}{2} \cdot 4 \cdot 6 = 12$. Hypotenuse $= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$. Distance d satisfies $\frac{1}{2} \cdot 2\sqrt{13} \cdot d = 12$, so $d = \frac{12}{\sqrt{13}}$.

Full method - how and why

Perpendicular distance from a point to a line. The 'area two ways' method avoids memorising the distance formula.

- 1 The line cuts the axes at $(4, 0)$ and $(0, 6)$, so with the origin it forms a right-angled triangle of area $\frac{1}{2} \times 4 \times 6 = 12$.

why: The two intercepts and the origin make a triangle whose legs lie along the axes - its area is easy.

- 2 The hypotenuse (the segment of L) has length $\sqrt{4^2 + 6^2} = \sqrt{52} = 2\sqrt{13}$.

why: Pythagoras on the same triangle.

- 3 Compute the area again using the hypotenuse as base and the required distance d as height: $\frac{1}{2} \times 2\sqrt{13} \times d = 12$.

why: The perpendicular distance from the origin to L is the height of that triangle on that base.

- 4 Solve: $d = \frac{12}{\sqrt{13}}$.

why: Equivalently, the formula $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$ with $3x + 2y = 12$ gives $\frac{12}{\sqrt{13}}$.

Where it goes wrong

- Using the distance to one of the intercepts $(4, 6 \text{ or } 2\sqrt{13})$ instead of the perpendicular distance.
- Forgetting the square root and answering $\frac{12}{13}$.

Takeaway Area computed two ways turns a distance problem into arithmetic; or use $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

Question 9 LIN-04 · Straight lines · spec MM3.1

The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

Answer: B

Intersection: $2x + 1 = 7 - x \Rightarrow x = 2$, $y = 5$. x -intercepts: $-\frac{1}{2}$ and 7 , base $= \frac{15}{2}$. Area $= \frac{1}{2} \cdot \frac{15}{2} \cdot 5 = \frac{75}{4}$.

Full method - how and why

A triangle bounded by two lines and an axis. Find the three vertices, then use the axis segment as the base.

- 1 Find where the two lines meet: $2x + 1 = -x + 7 \Rightarrow 3x = 6 \Rightarrow x = 2$, $y = 5$. Apex $(2, 5)$.

why: Equate the two expressions for y .

- 2 Find the x -intercepts: $2x + 1 = 0$ gives $x = -\frac{1}{2}$; $-x + 7 = 0$ gives $x = 7$.

why: The third side is along the x -axis, so the other two vertices are the x -intercepts.

- 3 Base length: from $-\frac{1}{2}$ to 7 is $7 - (-\frac{1}{2}) = \frac{15}{2}$.

why: Subtract, do not add; the negative intercept makes the base longer, not shorter.

- 4 Height is the y -coordinate of the apex, 5 : area $= \frac{1}{2} \times \frac{15}{2} \times 5 = \frac{75}{4}$.

why: With a horizontal base, the height is just the apex's distance from the axis.

Where it goes wrong

- Using base $7 - \frac{1}{2} = \frac{13}{2}$ by mishandling the negative intercept.
- Using the apex x -coordinate 2 as the height.

Takeaway Put the base on an axis whenever you can: the height is then simply a coordinate.

Question 10 LIN-08 · Straight lines · spec MM3.1

For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?

Answer: A

Gradients $-\frac{k}{3}$ and $\frac{2}{k-1}$. Perpendicular when the product is -1 : $\frac{-2k}{3(k-1)} = -1 \Rightarrow 2k = 3k - 3 \Rightarrow k = 3$.

Full method - how and why

A perpendicularity condition with the parameter in both equations. Write both gradients, multiply, set to -1 .

- 1 First line $kx + 3y = 2$ has gradient $-\frac{k}{3}$.
why: Using $-\frac{a}{b}$ avoids rearranging.
- 2 Second line $2x - (k - 1)y = 5$ has gradient $-\frac{2}{-(k-1)} = \frac{2}{k-1}$.
why: Here $b = -(k - 1)$, so the two minus signs cancel.
- 3 Set the product to -1 : $-\frac{k}{3} \times \frac{2}{k-1} = -1$, i.e. $\frac{-2k}{3(k-1)} = -1$.
why: Perpendicular lines have gradients multiplying to -1 .
- 4 Solve: $-2k = -3(k - 1) \Rightarrow -2k = -3k + 3 \Rightarrow k = 3$.
why: Check: gradients -1 and 1 , product -1 ✓.

Where it goes wrong

- Mishandling the sign of b in the second line and getting $k = -3$ - which is offered.
- Setting the product to $+1$ or the gradients equal (that would be the parallel condition).

Takeaway Read gradients as $-\frac{a}{b}$ straight from $ax + by = c$, watching the sign of b carefully.

Question 11 LIN-09 · Straight lines · spec MM3.1

The point P lies on the line $y = 2x - 1$ and is equidistant from $(0, 0)$ and $(4, 2)$. Find the coordinates of P .

Answer: B

Equidistant means P is on the perpendicular bisector of $(0, 0)$ - $(4, 2)$: midpoint $(2, 1)$, gradient -2 : $y = -2x + 5$.

With $y = 2x - 1$: $4x = 6$, $x = \frac{3}{2}$, $y = 2$. So $P = (\frac{3}{2}, 2)$.

Full method - how and why

'Equidistant from two points' means 'on their perpendicular bisector'. Intersect that with the given line.

- 1 Find the perpendicular bisector of $(0, 0)$ and $(4, 2)$: midpoint $(2, 1)$, segment gradient $\frac{1}{2}$, so bisector gradient -2 .
why: Equidistant points form the perpendicular bisector - far quicker than setting two distance formulas equal and squaring.
- 2 Its equation: $y - 1 = -2(x - 2)$, i.e. $y = -2x + 5$.
why: Point-gradient form, then tidy.
- 3 Solve simultaneously with $y = 2x - 1$: $2x - 1 = -2x + 5 \Rightarrow 4x = 6 \Rightarrow x = \frac{3}{2}$.
why: Equate the two expressions for y .
- 4 Then $y = 2(\frac{3}{2}) - 1 = 2$, so $P = (\frac{3}{2}, 2)$.
why: Check: distance to $(0, 0)$ is $\sqrt{\frac{9}{4} + 4}$ and to $(4, 2)$ is $\sqrt{\frac{25}{4} + 0}$ - both $\frac{5}{2}$ ✓.

Where it goes wrong

- Using the midpoint $(2, 1)$ as the answer without checking it lies on $y = 2x - 1$ (it does not).
- Algebra slips when squaring distances, if you take that longer route.

Takeaway 'Equidistant from two points' $\Rightarrow$ perpendicular bisector; then intersect with whatever else is given.

Question 12 LIN-11 · Straight lines · spec MM3.1

The vertices of a triangle are $A(0, 0)$, $B(6, 0)$ and $C(2, 4)$. Find the equation of the line through C that bisects the area of the triangle.

Answer: F

A line through a vertex bisects the area iff it passes through the midpoint of the opposite side, $(3, 0)$. Through $(2, 4)$ and $(3, 0)$: gradient -4 , $y = -4x + 12$.

Full method - how and why

A line through a vertex bisects a triangle's area exactly when it hits the midpoint of the opposite side - the two halves then have equal bases and the same height.

1 Identify the side opposite C : it is AB , from $(0, 0)$ to $(6, 0)$.

why: The line must pass through C , so it cuts the opposite side.

2 Its midpoint is $(3, 0)$.

why: A median divides the triangle into two triangles with equal bases (3 each) and the same apex height, hence equal areas.

3 Find the line through $C(2, 4)$ and $(3, 0)$: gradient $= \frac{0 - 4}{3 - 2} = -4$.

why: Two points give the gradient directly.

4 Equation: $y - 0 = -4(x - 3)$, i.e. $y = -4x + 12$.

why: Using the simpler point $(3, 0)$ keeps the arithmetic clean.

Where it goes wrong

- Using the midpoint of the wrong side, or the centroid.
- Sign slip giving $y = 4x - 4$, which passes through $(1, 0)$ - an offered option.

Takeaway A median (vertex to opposite midpoint) always bisects the area, whatever the triangle's shape.